# Practice workbook · Write a logic statement for a problem

Turning a written set of conditions into a Boolean expression using AND, OR, NOT and brackets.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-11

# 1. Question 5 · part d.ii · 3 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

**5** Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

**(2)**

| Requirement | Example |
|---|---|
| Input | |
| Process | Calculate the path the bee will move along to its new position |
| Output | |

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# 1. Question 5 · part d.ii · 3 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

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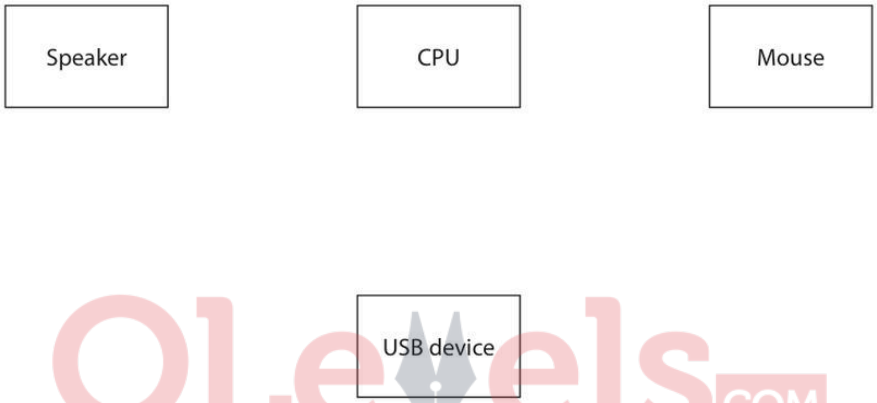
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(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

**(3)**




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# 1. Question 5 · part d.ii · 3 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

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# 1. Question 5 · part d.ii · 3 marks

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Worked-example retry · Attempt part d.ii only.

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(c) A washing machine uses several different embedded systems.

One embedded system uses a switch to identify the type of wash cycle selected by the user.

Describe **one other** example of an embedded system found in a washing machine.

**(2)**

(d) Programs use logic statements to control physical hardware.

(i) A window shuts when the temperature is too cool or it is a rainy night.

The values are defined as:

- A shows it is night time
- B shows it is too cool
- C shows it is raining.

Complete the truth table to show the results of each operation.

Two rows have been done for you.

**(3)**

| A | B | C | A AND C | (A AND C) OR B |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | 1 | 1 |

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P 7 2 9 3 7 A 0 1 7 2 4

17

*Turn over* ▶

# 1. Question 5 · part d.ii · 3 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

(ii) A warehouse has an automated alarm system.

When the alarm system is activated it will sound if:

- a movement sensor (M) is activated
- a pressure pad (P) is activated
- a key code (C) to deactivate the alarm system has not been entered.

Construct a logic statement, using AND, OR and NOT with the letters M, P and C, to show the conditions that will sound the alarm.

**(3)**

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# 1. Question 5 · part d.ii · 3 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

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(e) A computer with a single CPU runs several processes at the same time.

This computer is multitasking.

Describe how the operating system enables processes to share a single CPU.

(2)

**(Total for Question 5 = 18 marks)**

Working and answer

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## 2. Question 5 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

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**5** Programmers work as part of a team to develop software.

(a) They use truth tables to determine if an expression evaluates to true or false.

Complete this truth table for the Boolean expression

**P AND (Q OR R)**

You may not need to use all rows.

**(3)**

| P | Q | R | Q OR R | P AND (Q OR R) |
|---|---|---|---|---|
| | | | | |
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## 2. Question 5 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

(b) The team is working on software for a ride-on lawn mower.

The lawn mower must not start unless it is safe for the driver.

- The brake (B) must be engaged.
- The driver's seat sensor (S) must be engaged.
- The blade safety lock (L) must be removed.

Construct a Boolean expression, using AND, OR, and NOT with the letters B, S and L to show the safe starting condition.

**(2)**

(c) The team uses code reviews as part of the software development process.

State **two** ways in which code reviews can help protect software systems from cyber attacks.

**(2)**

1 .....

2 .....

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## 2. Question 5 · part b · 2 marks · Working and answer

Working and answer

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## 3. Question 6 · part c · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

**6** Akiko develops software programs.

(a) Akiko can use a low-level or a high-level programming language to write code.

(i) Give **one** benefit of using a low-level programming language to develop a program.

**(1)**

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(ii) Give **one** drawback of using a low-level programming language to develop a program.

**(1)**

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(b) Akiko is writing an anti-virus program.

(i) The program will include a virus removal function.

Give **two other** functions that should be included.

**(2)**

1 .....

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2 .....

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(ii) An anti-virus program is one type of anti-malware.

Give **one other** type of anti-malware.

**(1)**

.....

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## 3. Question 6 · part c · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

(c) Akiko is also working on a burglar alarm system.

The alarm must only be triggered if:

- the alarm (A) is set
- it is dark outside (O)
- a window (W) or a door (D) has been opened.

Construct a Boolean expression, using AND, OR and NOT with the letters A, O, W, and D to show the conditions that will trigger the alarm.

**(4)**

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## 3. Question 6 · part c · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

(d) Akiko wants to know how the software she writes can be protected.

Discuss how the software can be protected from illegal copying and/or illegal distribution.

You should consider:

- intellectual property
- licensing.

**(6)**

**(Total for Question 6 = 15 marks)**

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## 3. Question 6 · part c · 4 marks · Working and answer

Working and answer

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# 4. Question 5 · part e · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part e only.

**5** A team is developing a computer game.

(a) Complete the table by adding a tick (✓) to show the characteristics of program language translators.

You may select more than one translator per characteristic.

**(3)**

| Characteristic | Translator | | |
|---|---|---|---|
| | **Assembler** | **Interpreter** | **Compiler** |
| Does **not** have to be present in memory to execute the translated code. | | | |
| Translates high level language source code. | | | |
| Executes high level code until either the program has finished, or a syntax error is encountered. | | | |
| Translates low level language source code into machine code. | | | |

(b) Luca is testing the computer game.

The computer that he is using is an older model with a processor that runs at a low clock speed.

Explain **one** drawback of testing the game using this computer.

**(2)**

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P 7 5 7 3 6 A 0 1 8 2 8

## 4. Question 5 · part e · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part e only.

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(c) Anna uses utility software.

Complete the table by adding the type of utility software she should use to solve each issue.

| Issue | Utility software |
|---|---|
| Her word processor cannot read all of a file because it is corrupt. | |
| She has a JPEG image and needs to send it to a publisher who can only read PNG image files. | |
| There has been a flood and files on a server have been damaged, but Anna needs to access her data. | |
| Her hard drive is running very slowly when she tries to access a file. | |

(4)

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# 4. Question 5 · part e · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part e only.

(d) Tarik is writing an instruction manual for the computer game.

He decides to use software to compress the instruction manual before emailing it to a proofreader for checking.

State and justify the most appropriate compression method.

(3)

Method

.....

Justification

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(e) Members of the team developing the game are either coders or designers.

Coders (C) write the application code.

Designers (D) design the graphics for the game and write instruction manuals.

Team members have one role only.

Construct a logic statement, using AND, OR and NOT with the letters C and D, to show the conditions for being a member of the team.

(3)

**(Total for Question 5 = 15 marks)**

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## 4. Question 5 · part e · 3 marks · Working and answer

Working and answer

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